Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam E · Form 1: Constraint analysis

Multiple Choice Questions: Select the best answer. (1 point each)

1. A ramp is tilted slowly. A block first slips at θc. Which surface parameter can be inferred directly?

(A) μk=tanθc

(B) μk=sinθc

(C) μs=cosθc

(D) μs=tanθc

2. A 2 kg cart on a frictionless horizontal plane experiences 12 N east and 13 N north. Find the magnitude of a third force that would maintain constant velocity.

(A) -17.692 N

(B) 35.384 N

(C) 8.846 N

(D) 17.692 N

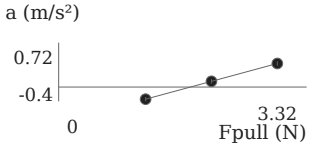
3. A 2 kg cart slides right against constant 2 N friction to the left. Fpull (N) -> a (m/s²): (1.32, -0.34); (2.32, 0.16); (3.32, 0.66). At fixed sensor resolution, how can slope uncertainty be reduced?

(A) Use a wider horizontal baseline within the valid regime.

(B) Discard the sign of every reading.

(C) Shorten the baseline.

(D) Append digits to one reading.



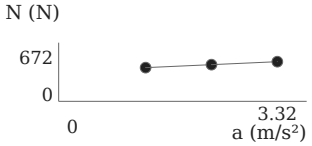
4. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². Double passenger mass in the same gravity. At the same X=2.32, what is the new minus old vertical reading?

(A) 647.5

(B) 616

(C) 626.5

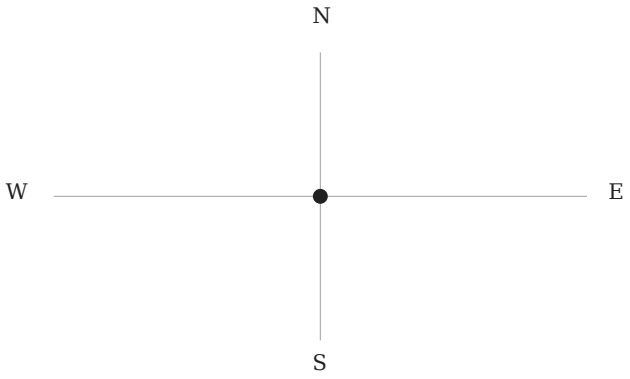
(D) 595



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two students push a cart of mass 16 kg on a frictionless horizontal floor. A exerts 340 N at 30° north of east. B exerts 460 N at 42° south of west.

a. Draw and label the actual forces acting in the plane of the floor on the dot. Draw whole vectors, not components. **(1 pt)**



b. Show net-force x and y components, then find acceleration magnitude and direction using compass words. **(3 pt)**

6. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². sinθ -> a (m/s²): (0.1584, 1.584); (0.2784, 2.784); (0.3984, 3.984).

a. Build the model for ramp tilt experiment from physics and predict both slope and intercept. **(2 pt)**

b. Double block mass on the same frictionless ramp. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. Hanging m1 connects by an ideal rope to m2 on an incline θ. A vertically downward F_A acts on m2, which slides uphill. Given μ_k, derive acceleration with uphill positive for m2. **(3 pt)**